Given a temperature of 201 K, convert to degrees Fahrenheit.

A metal has a coefficient of linear expansion of 2.6×10^{-5} $(C°)^{-1}$. Suppose a pipe made of this metal is 1.7 m long. Calculate the temperature increase in C° if its length increases by 7.4 mm.

A metal has a coefficient of linear expansion of 1.1×10^{-5} $(C°)^{-1}$. Suppose a pipe made of this metal is 5.2 m long. Calculate its length increase in mm if it is warmed by 46 C°.

ΔL = _____mm

Suppose a glass bottle with volume 0.012 m^3 if filled to the brim with a fluid of unknown coefficient of volume expansion. When warmed by 7.1 C°, 16 mL spill onto the lab table. If the glass has linear coefficient 1.9×10^{-5} K^{-1}, calculate the unknown volume coefficient of the fluid in $(kC°)^{-1}$.

(Be careful with units and note that 1000 C° = 1 kC°.)

$\beta =$ _____ $(kC°)^{-1}$

A bottle filled with 1.3-L of water falls from a height of 150.2 m and hits the ground ($g = 9.8$ m/s^2). If all the mechanical energy loss of the water went into heating the water, find its **change** in temperature (in C°).

Suppose 3,141 J of energy are transferred to a block of mass 4.3 kg and specific heat 645 J/(kg·C°). Calculate the temperature change in Celsius degrees.

Suppose transferring 24,512 J to a substance of mass 2.8 kg raises the temperature by 11 C°. Calculate the specific heat of the substance in J/(kg·C°).

Suppose 5.6 kg of chloroform at 13.5°C is poured into 9.5 kg of propylene glycol at 36.8°C. Calculate the final equilibrium temperature, neglecting any energy processes associated with mixing, given c_{prop} = 2500 J/(kg·K) and $c_{chloroform}$ = 1050 J/(kg·K).

T = _____ °C

An ice cube tray holds 0.29 kg of water at 17°C. How much energy in kJ much be removed from the water to freeze it?

[Use: L_f = 335,000 J/kg; L_v = 2,260,000 J/kg; c = 4190 J/(kg·K).]

How much energy (in kJ) is required to convert 2.69 kg of ice at -20°C to water at 16°C?

[Use: c_{ice} = 2100 J/(kg·K); c_{water} = 4190 J/(kg·K); $L_{f,water}$ = 334x10^3 J/kg]

Suppose 7.2 kg of ice at 0°C is placed in 11.5 kg of water at 28.4°C. Calculate the final equilibrium temperature in Celsius. L_f = 3.35 x 10^5 J/kg; c = 4190 J/(kg·K).

[Hint: There are two possible outcomes - the ice will be either fully or partially melted. Start by figuring out which occurs in this case.]

A window has dimensions 1.6 m × 2 m × 13.9 mm where the latter quantity is the thickness in millimeters. The temperature outside is -17°C, and inside it is 18°C. If the thermal conductivity of the glass is 0.8 W/(m·K), calculate the magnitude of the energy transfer rate through the window in kW.

H = _____kW

A ball hanging by a wire has a radius of 3.1 m and emissivity 0.63. If the surface temperature of the ball is 92°C and the room is at temperature 28°C, calculate the net rate in kW at which energy is radiated by the ball.

H = _____kW